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## What is File Handling?
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**** File handling is used to create, read, write, update and delete files using Python Program.**

**** A file is used to store data permanently.**

**** Normal variables store data temporarily.**

**** Files store data permanently in system storage.**

Example :

```
-- Student records
-- Employee details
-- Course details
-- Logs
-- Reports
-- Configuration data
```

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## Types of Files
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Python mainly works with two types of files:

1) Text Files

2) Binary Files

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## Text Files
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Text files store data in readable text format.

Examples :

```
.txt
.csv
.json
.py
.html
```

Example :

```
students.txt
courses.txt
employees.csv
```

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## Binary Files
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Binary files store data in binary format.

Examples :

Images
Videos
Audio files
PDF files

Example :

photo.jpg
video.mp4
resume.pdf

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## File Handling Steps
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File handling usually contains 3 steps:

Step-1 : Open File / Create File

Step-2 : Perform Operation

Step-3 : Close File

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## open() Function
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1) open () : It is used to open a file

Syntax : open("file_name", "mode")

=> File modes are used to specify the operation.

r - read
w - write
a - append
x - create

r+ - read and write mode
w+ - write and read mode
a+ - append and read mode

rb - read binary file

wb - write binary file

ab - append binary file

read () : It reads complete file data at a time.

read(number) : reads specific number of characters.

readline() : It reads one line at a time

readlines() : It reads all lines and returns a list

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## write ( ) :
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## writelines( ) :
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Reading file using for loop
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```
file = open("students.txt", "r")
```

```
for line in file:
    print(line)
```

```
file.close()
```

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## write() Method
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write() is used to write data into file.

Example :

```
file = open("courses.txt", "w")
```

```
file.write("Python\n")
file.write("Java\n")
file.write("DevOps\n")
```

```
file.close()
```

Output :

courses.txt file will contain course names.

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## writelines() Method
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writelines() is used to write multiple lines into file.

Example :

```
courses = ["Python\n", "Java\n", "DevOps\n"]
```

```
file = open("courses.txt", "w")
```

```
file.writelines(courses)
```

```
file.close()
```

Output :

courses.txt file will contain all courses.

Important:

writelines() does not add new line automatically.

We should add \n manually.

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## close() Method
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close() is used to close the file.

Example :

```
file = open("students.txt", "r")

data = file.read()

print(data)

file.close()
```

Important:

Always close the file after file operation.

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## Why close() is Important?
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close() is important because:

- It releases system resources.
- It saves pending data into file.
- It avoids file corruption.
- It is a good programming practice.

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## File Handling Using "with"
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**** with statement is the best way to handle files.**

**** It automatically closes file after operation.**

Syntax :

```
with open("filename", "mode") as file:
    file operation
```

Example :

```
with open("students.txt", "r") as file:
    data = file.read()
    print(data)
```

```
with open("courses.txt", "w") as file:
    file.write("JAVA\n")
    file.write("PYTHON\n")
    file.write("GEN AI\n")
```

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## Check File Exists
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We can check whether file exists or not using os module.

Example :

```
import os

if os.path.exists("students.txt"):
    print("File exists")
else:
    print("File not found")
```

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## Delete File
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We can delete file using os.remove().

Example :

```
import os

if os.path.exists("students.txt"):
    os.remove("students.txt")
    print("File deleted")
else:
    print("File not found")
```

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## Rename File
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We can rename file using os.rename().

Example :

```
import os

os.rename("students.txt", "student_details.txt")

print("File renamed successfully")
```

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## Create Folder
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We can create folder using os.mkdir().

Example :

```
import os

os.mkdir("reports")

print("Folder created successfully")
```